Business Area Computation: Mathematical Documentation

1 Business Area From POI Clusters (Geometry Pipeline)

This section defines a purely geometric business area using business-related POIs. The pipeline is: (i) project coordinates to a local metric CRS, (ii) restrict analysis to a core region around the city center, (iii) retrieve business POIs, (iv) cluster them with DBSCAN, and (v) convert the dominant cluster to a polygon whose area and centroid define the business area and business center.

Coordinate system and core region

We work with a city boundary polygon B and two reference points: a geocoded city center c and a historic center g, both initially in lon-lat.

$$\textbf{Inputs: } B \subset \mathbb{R}^2,\ c = (\lambda_c, \varphi_c),\ g = (\lambda_g, \varphi_g),\ R \text{ (km)}$$

A UTM zone is chosen from longitude, and the EPSG code is determined by hemisphere:

$$z = \left\lfloor \frac{\lambda + 180}{6} \right\rfloor + 1$$

$$\text{EPSG} = \begin{cases} 32600 + z, & \varphi \geq 0, \\ 32700 + z, & \varphi < 0 \end{cases}$$

Let T be the projection from lon-lat into the chosen local metric CRS:

$$T:\ (\lambda, \varphi) \mapsto (x, y) \in \mathbb{R}^2$$

We define a circular core region of radius $1000R$ meters around the projected city center, and intersect it with the projected boundary:

$$C = \left\{ u \in \mathbb{R}^2 :\ \|u - T(c)\| \leq 1000R \right\}$$
$$K = T(B) \cap C$$

If the boundary intersection fails (empty or zero area), we fall back to the core circle:

$$K = \begin{cases} K, & \text{Area}(K) > 0, \\ C, & \text{Area}(K) = 0 \end{cases}$$

POI clustering and polygon construction

Let $\mathcal{S}$ be the set of OSM tags considered "business" (shops, offices, etc.). The metric POI point set is:

$$X = \left\{ T(p)\ :\ p \in \text{OSM},\ p \in T^{-1}(K),\ \text{tag}(p) \in \mathcal{S} \right\}$$

DBSCAN is applied to X using a metric radius ε and minimum samples m. The neighborhood and core condition are:

$$N_\varepsilon(x_i) = \{x_j \in X :\ \|x_j - x_i\| \leq \varepsilon\}$$

$$x_i \text{ is core} \iff |N_\varepsilon(x_i)| \geq m$$

DBSCAN assigns labels:

$$\ell_i \in \{-1, 0, 1, 2, \dots\}, \qquad (\ell_i = -1 \text{ means noise})$$

We keep the dominant (largest) non-noise cluster:

$$k^\star = \arg\max_{k \geq 0} |\{i : \ell_i = k\}|$$
$$X^\star = \{x_i \in X : \ \ell_i = k^\star\}$$

To obtain a continuous business footprint, we buffer each cluster point by radius r and take the union:

$$\mathcal{B}(x, r) = \left\{u \in \mathbb{R}^2 : \ \|u - x\| \leq r\right\}$$
$$U = \bigcup_{x \in X^\star} \mathcal{B}(x, r)$$

If U yields multiple polygons, we keep the largest component:

$$P = \text{LargestPolygon}(U)$$

Optionally, the polygon is simplified with tolerance s:

$$P_s = \begin{cases} \text{Simplify}(P, s), & s > 0, \\ P, & s \leq 0 \end{cases}$$

The business area and business center are then:

$$A_{\text{km}^2} = \frac{\text{Area}(P_s)}{10^6}$$
$$\mu = \text{Centroid}(P_s)$$

If a valid polygon cannot be produced, we define a fallback:

$$\mu = T(c), \qquad A_{\text{km}^2} = 0$$

Finally, the separation from the historic center is computed in the metric CRS:

$$g_m = T(g)$$
$$d = \|\mu - g_m\| = \sqrt{(\mu_x - g_x)^2 + (\mu_y - g_y)^2}$$

2 Model-Based Shares and Fixed-Point Business Center

This section defines a model that assigns each grid cell a business "share" (a value in $[0, 1]$) based on a logistic specification, and then solves for a business center μ that is consistent with those shares. Consistency is needed because travel time to the center depends on μ, but shares also determine μ through a weighted average. This circular dependence is resolved by fixed-point iteration.

Cells, features, and standardization

Let grid cells be indexed by $i = 1, \ldots, N$, with metric centroids $c_i = (c_{x,i}, c_{y,i}) \in \mathbb{R}^2$. Each cell has observed features:

$$\begin{aligned} d_i^{(g)} &= \texttt{dist_to_g_m}, & h_i &= \texttt{herit_density_500m}, \\ p_i &= \texttt{pop_density_per_km2}, & \ell_i &= \texttt{log_rent}, \\ r_i &= \texttt{rent_missing}. \end{aligned}$$

Two interactive multipliers are applied: transport factor t and historic-amenity strength a:

$$t \in [0.1, 2.0], \qquad a \in [0.1, 2.0]$$

Let v be the travel speed in meters per minute:

$$v = \text{speed_m_per_min}$$

Each feature is standardized using training means and standard deviations:

$$z_x(i) = \frac{x_i - \mu_x}{\sigma_x}, \qquad \sigma_x \neq 0$$

Travel time depends on the unknown center

For a candidate business center $\mu = (\mu_x, \mu_y)$, define distance and travel time:

$$\begin{aligned} d_i(\mu) &= \|c_i - \mu\| = \sqrt{(c_{x,i} - \mu_x)^2 + (c_{y,i} - \mu_y)^2} \\ \tau_i(\mu, t) &= \frac{d_i(\mu)}{v\, t} \end{aligned}$$

Standardized regressors used in the logit include standardized travel time and standardized cell attributes:

$$\begin{aligned} z_\tau(i; \mu, t) &= \frac{\tau_i(\mu, t) - \mu_\tau}{\sigma_\tau} \\ z_g(i) &= \frac{d_i^{(g)} - \mu_g}{\sigma_g}, \qquad z_h(i) = \frac{h_i - \mu_h}{\sigma_h} \\ z_p(i) &= \frac{p_i - \mu_p}{\sigma_p}, \qquad z_\ell(i) = \frac{\ell_i - \mu_\ell}{\sigma_\ell}, \qquad z_r(i) = \frac{r_i - \mu_r}{\sigma_r} \end{aligned}$$

Logit model for shares

The linear predictor is:

$$\eta_i(\mu, t, a) = \beta_0 + \beta_\tau z_\tau(i; \mu, t) + \beta_g\, (a\, z_g(i)) + \beta_h\, (a\, z_h(i))$$

$$+ \beta_p z_p(i) + \beta_\ell z_\ell(i) + \beta_r z_r(i)$$

For numerical stability, the predictor may be clamped to a bounded interval:

$$\tilde{\eta}_i(\mu, t, a) = \min\{35, \max\{-35, \eta_i(\mu, t, a)\}\}$$

Shares are computed with the logistic function:

$$s_i(\mu, t, a) = \frac{1}{1 + \exp\big(-\tilde{\eta}_i(\mu, t, a)\big)}$$

Fixed-point definition of the business center

Given shares, the implied business center is a weighted average of cell centroids:

$$S(\mu, t, a) = \sum_{i=1}^{N} s_i(\mu, t, a)$$

$$F(\mu; t, a) = \frac{1}{S(\mu, t, a)} \sum_{i=1}^{N} s_i(\mu, t, a)\, c_i$$

A fixed point $\mu^\star$ satisfies $\mu^\star = F(\mu^\star; t, a)$. We approximate it by iteration starting from an initial guess $\mu^{(0)} = (\mu_{0x}, \mu_{0y})$:

$$\mu^{(k+1)} = F(\mu^{(k)}; t, a)$$

$$\left\| \mu^{(k+1)} - \mu^{(k)} \right\| < \delta \quad \text{(stop)}$$

The outputs are:

$$\mu^\star = \mu^{(K)}, \qquad s_i^\star = s_i(\mu^\star, t, a)$$

Separation from the historic center $g = (g_x, g_y)$ in meters:

$$\text{sep}_m = \|\mu^\star - g\|$$

Optionally, for compact transport to the browser, shares can be quantized:

$$b_i = \text{round}(255 \cdot \min\{1, \max\{0, s_i^\star\}\}) \in \{0, \ldots, 255\}$$

Business polygon from shares

To convert the share field into a single business polygon, we select a subset of high-share cells that captures a target fraction of total share mass.

Total mass:

$$M = \sum_{i=1}^{N} s_i^\star$$

Let $s_{(1)}^\star \geq s_{(2)}^\star \geq \cdots \geq s_{(N)}^\star$ be sorted shares, and $\pi(k)$ the index of the kth largest share:

$$\pi(k) = \text{index of } s_{(k)}^\star$$

With configuration parameters α (target mass fraction), $K_{\max}$, $K_{\min}$:

$$\alpha = \text{target_mass}, \qquad K_{\max} = \text{max_points}, \qquad K_{\min} = \text{min_points}$$

Select the smallest K that meets the target mass (subject to min and max):

$$K = \min \left\{ k \le K_{\max} : \; k \ge K_{\min} \;\wedge\; \sum_{j=1}^{k} s^{\star}_{\pi(j)} \ge \alpha M \right\}$$

$$\mathcal{I} = \{\pi(1), \ldots, \pi(K)\}, \qquad P = \{c_i : \; i \in \mathcal{I}\}$$

Define radii in meters:

$$r = \text{buffer_km} \cdot 1000, \qquad \rho = \text{close_km} \cdot 1000, \qquad \kappa = \text{corridor_km} \cdot 1000$$

Create a buffered union of the selected points:

$$U = \bigcup_{i \in \mathcal{I}} \mathcal{B}(c_i, r)$$

Optionally apply a morphological closing (buffer then unbuffer) to smooth small gaps:

$$U^{\text{cl}} = (U \oplus \mathcal{B}(0, \rho)) \ominus \mathcal{B}(0, \rho)$$

Let U^{cl} have connected components $Q_1, \ldots, Q_J$:

$$U^{\text{cl}} = \bigcup_{j=1}^{J} Q_j, \qquad Q_j \cap Q_{j'} = \varnothing \; (j \ne j')$$

Compute component masses by summing shares of points contained in each component:

$$m_j = \sum_{i \in \mathcal{I}} s^{\star}_i \, \mathbf{1}\{c_i \in Q_j\}$$

Discard tiny components using a minimum mass fraction γ:

$$\gamma = \text{min_component_mass}, \qquad \mathcal{J} = \left\{ j : \; m_j \ge \gamma \sum_{u=1}^{J} m_u \right\}$$

Let $i^{\star} = \pi(1)$ be the peak-share index and $c^{\star} = c_{i^{\star}}$ its location. Choose an anchor component, preferring the one containing the peak point:

$$j^{\star} = \begin{cases} \arg\{j \in \mathcal{J} : \; c^{\star} \in Q_j\}, & \exists j \in \mathcal{J} \text{ with } c^{\star} \in Q_j, \\ \arg\max_{j \in \mathcal{J}} m_j, & \text{otherwise} \end{cases}$$

Initialize $W_0 = Q_{j^{\star}}$. For remaining kept components, connect them to the current polygon by adding a corridor around the centroid-to-centroid segment:

$$\bar{w} = \text{Centroid}(W), \qquad \bar{q}_j = \text{Centroid}(Q_j)$$

$$W \leftarrow W \;\cup\; Q_j \;\cup\; \Big(\text{Seg}(\bar{w}, \bar{q}_j) \oplus \mathcal{B}(0, \kappa)\Big), \qquad j \in \mathcal{J} \setminus \{j^{\star}\}$$

Optionally close again:

$$W^{\text{cl}} = (W \oplus \mathcal{B}(0, \rho)) \ominus \mathcal{B}(0, \rho)$$

Business area:

$$A_{\text{km}^2} = \frac{\text{Area}(W^{\text{cl}})}{10^6}$$

Finally, if desired for display, simplify in lon-lat after inverse projection:

$$\widetilde{W} = \text{Simplify}\big(T^{-1}(W^{\text{cl}}), \; \text{simplify_tol}\big)$$

3 External Resources and Links

Data sources (datasets)

- WorldPop (Armenia 2020 population raster, 100m): `https://data.worldpop.org/GIS/Population/Individual_countries/ARM/Armenia_100m_Population/ARM_ppp_v2c_2020.tif`
- OpenStreetMap (data used via OSMnx): `https://www.openstreetmap.org/`
- Nominatim (geocoding service): `https://nominatim.org/`
- Nominatim usage policy: `https://operations.osmfoundation.org/policies/nominatim/`
- Overpass API (public endpoint): `https://overpass-api.de/api/interpreter`
- Kaggle candidates:
  - `https://www.kaggle.com/datasets/molchoun/real-estate-data-armenia`
  - `https://www.kaggle.com/datasets/mirzoyanvahe/armenian-real-estate-market-data`
  - `https://www.kaggle.com/datasets/puff1n/armenian-flats-from-listam`

Map tile source (Leaflet)

- CARTO raster tiles (Positron / light_all): `https://\{s\}.basemaps.cartocdn.com/light_all/\{z\}/\{x\}/\{y\}\{r\}.png`

Other external links (libraries/assets)

- Leaflet CSS: `https://unpkg.com/leaflet@1.9.4/dist/leaflet.css`
- Leaflet JS: `https://unpkg.com/leaflet@1.9.4/dist/leaflet.js`
- Plotly: `https://cdn.plot.ly/plotly-2.30.0.min.js`
- Proj4js: `https://cdnjs.cloudflare.com/ajax/libs/proj4js/2.9.0/proj4.js`
- Pako: `https://cdn.jsdelivr.net/npm/pako@2.1.0/dist/pako.min.js`
- Scrollama: `https://unpkg.com/scrollama`
- Google Fonts: `https://fonts.googleapis.com` , `https://fonts.gstatic.com`
- Inter (font stylesheet variants):
  - `https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap`
  - `https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700;800;900&display=swap`
  - `https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800;900&display=swap`
- SVG namespace reference: `http://www.w3.org/2000/svg`

Publishing/deployment API link

- GitHub API: `https://api.github.com`